ECHO-7

Visual bay finding and docking | Product & operating guide

DE1-SoC / OV7670 / UGV02

Choose a blue bay marker and start a task. ECHO-7 searches, confirms the marker, aligns, approaches and parks, with live VGA feedback, remote stop and front sonar protection.

OV7670 camera → Live colour image

DE1-SoC FPGA → Cyclone V 5CSEMA5F31C6 | vision + control

UGV02 chassis → Differential-wheel motion

Also fitted: J3 RCWL-1601 sonar, NEC IR receiver/remote, VGA monitor, switches, keys, LEDs and six-digit display. USB-Blaster provides PC download/debug. Chassis UART: 115200 8N1, 3.3 V logic, common ground.

THREE OPERATING MODES

MODE / DISPLAY CODE	PURPOSE
0 - INTEGRATED	Camera-guided driving. Choose one bay; POWER or red2 starts a new task. Default after power-up/reset, with motion disarmed.
1 - VISION_ONLY	Live recognition, tracking, calibration and VGA; wheel commands stay zero. Use for marker and camera setup.
2 - CHASSIS_INJECT	PC/JTAG virtual targets test the same controller. Real IR and sonar stay active; loss of the PC injection session stops motion.

Choose modes with the JTAG debug tool. Changing mode stops the task. INTEGRATED needs no PC heartbeat.

RED AND BLUE MARKERS

Blue 1-6: destination bays. The matching switch selects the number to follow. Keep the complete border and white anchor visible.

Red1: colour calibration. Show red1 with its paired blue reference stripe, steadily. Motion pauses during calibration; wait for **CAL:Y**. The stripe is labelled **BJ**; it is not a bay.

Red2: reselect and start. Re-read the switches, relock and request a task. Move it out, then present it once beside the target. It does not choose a bay or restart an active task. Keep hands/cards clear of J3.

SWITCHES AND BUTTONS

SW0-SW5	SW0/1/2/3/4/5 select blue 1/2/3/4/5/6. **Exactly one ON.** No selection/multiple selections cannot start. Changing selection cancels motion.
SW6	Candidate colour/number diagnostics.
SW7	Centre colour sample and marker identity.
SW8	0: normal perspective handling. 1: bounding-box recognition bypass.
SW9	Camera-only display; cancels motion. OFF requires a new start to move again.
KEY0 / KEY1	KEY0: full reset. KEY1: next display page.

Remote POWER: start. **A:** latched emergency stop. After A, the first fresh POWER only releases the stop; release the key, then press POWER again to start. Holding POWER does not do both.

PC/JTAG motion tuning: VISION_ONLY only, disarmed and stopped. Settings and limits: README. Safety distance thresholds stay fixed.

WHAT THE VGA IMAGE SHOWS

Upright camera image, red/blue colour contours and accepted labels **R1/R2**, **B1-B6**, or **BJ**. **CAL:W/B/Y** = waiting/calibrating/calibrated. **FRAME:GOOD/BAD/STALE** reports image health. **STATE** is the calibration interaction state, not driving state.

SW6: RGB/HSV, colour, number or UNKNOWN, and rejection code E. PAGE:1/2 alternates about every 2 s across 16 slots; yellow sample marks identify slots.

SW7: white 8 x 8 centre sample with RGB/HSV and the identity of a complete marker covering it. **SW8:** BYPASS:1 changes recognition, not the camera view or safety. **SW9:** removes overlays.

READING THE SIX-DIGIT DISPLAY

page | mode | four data digits

d1 0520 = distance page, VISION_ONLY, **520 mm**. The second digit is the mode code above, not the bay number.

d	Decimal millimetres. **FAr** = valid out-of-range indication, not a measured 4 m. Four blank digits = invalid distance.
H	Last IR word, upper half in hexadecimal. Right pair **12** = POWER, **0F** = A (examples ED12 / F00F).
L	Last IR word, lower/address half in hexadecimal. H/L are blank until a complete valid IR word arrives.

KEY1: d - H - L - d. Left digit **n** = stable no-selection; **C** = stable multi-switch conflict, replacing the page letter. No ARM or driving-state page.

LEDs: 0 motion permitted; 1 IR stop latched; 3 docked; 8 target locked and fresh.

APPROACH / STOP Default speeds: 5 cm/s, 3 cm/s within 500 mm. A locked target within +/-30 pixels of centre allows straight approach and docks at <=350 mm; restart to move again. Stop at <=250 mm, invalid/stale range or IR A. Obstacle HOLD can release after three fresh >=300 mm readings. Target loss: fully stop for 1.5 s, then reconfirm from fresh frames before searching.

SEARCH / RESET An unseen target uses bounded low-speed search. After a search sighting: stop, wait and reconfirm. If the sighting is lost, up to three 150 ms reverse steps may recover it; each stops to reconfirm within the same scan budget. Search is approximate, not a guaranteed 360 degrees. Changing search speed requires new floor scan calibration. Reset/power loss restores tuning defaults and clears colour/scan calibration; restore the scan setting from the PC before unseen-target search. Visible, locked targets can still be approached.